like ovalbumin, conalbumin, ovamucoid, ovomucin and avidin. The protein ovomucin is responsible for the jelly-like character of egg white and thickness of the albumin. Avidin binds with biotin and makes the vitamin unavailable. But avidin is denatured by heat and thus cooked egg does not affect the availability of biotin.

Egg yolk comprises mostly 25-33 percent of fat and 15-17 percent protein and the remaining water. The major proteins in egg yolk are lipoproteins which include lipovitellins and lipovitellinin. These lipoproteins are responsible for the excellent emulsifying properties of egg yolk, when it is used in products such as mayonnaise.

4.4.2 The value of eggs in the diet

Eggs are an excellent and relatively cheap source of high biological value protein. Egg proteins have an excellent supplementary value to all other plant protein foods. Hence a combination of eggs with any of the cereal or cereal pulse mixture will enhance the protein quality of food. They also provide vitamins A, D, E and riboflavin. Egg yolk is a good source of carotene and iron. Egg is one of the richest sources of lecithin- a phospholipid which forms a part of the structure of every cell wall in the body. Egg also provides essential fatty acids like linoleic acid and arachidonic acid.

4.4.3 Evaluation of egg quality

Egg is an excellent food and hence its quality is of very great importance. Fresh eggs have the best quality. Quality of eggs can be determined by candling where the egg is held against a source of strong light. Candling will reveal

a) a crack in the shell.

b) the size of the air cell.

c) the firmness of the white.

d) the position of the yolk.

e) the presence of foreign substances.

Table 4.5 Nutritive value of Egg (per 100 g)

Food	Energy (Kcal)	Protein (g)	Fat (g)	Calcium (mg)	Iron (mg)
Egg (hen)	173	13.3	13.3	60	2.1

Table 4.6 Characteristics of fresh and deteriorated eggs

Characteristics	Fresh egg	Deteriorated egg
Consistency of Egg white	Firm and viscous	Thin and runny
Position of the yolk	Centre of the white	Yolk moves toward the shell
Air cell	Small	large
Chalazae	Strong	weakens
Vitelline membrane	Intact and strong	Weak, Ruptures- white mixes with the yolk

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